

Data & Population

This analysis uses four files from the 2023 NFL Big Data Bowl dataset covering Weeks 1–9 of the 2022 NFL season: `player_play.csv`, `plays.csv`, `players.csv`, and `games.csv`. Tracking files were downloaded but not used. All analysis is conducted at the player-play level, with `player_play.csv` serving as the primary file for metric computation.

Before any analysis, three play types were excluded. Plays nullified by penalty (`playNullifiedByPenalty == "Y"`), QB kneels (`qbKneel == 1`), and QB spikes (`qbSpike == TRUE`) were removed from the dataset. NA values in `qbSpike` were retained as non-spike plays. All analysis was further restricted to pass plays only (`isDropback == TRUE`), eliminating run plays from the population entirely.

The pass rusher population is defined by `wasInitialPassRusher == 1` in `player_play.csv` rather than by roster position. This captures all players who legitimately rushed the passer on a given play regardless of listed position, including 3-4 OLBs, nickel rushers, and any non-traditional pass rush alignments. Filtering by roster position would exclude legitimate pass rushers and introduce positional bias into the population.

Sacks are identified using `sackYardsAsDefense < 0` and half sacks using `halfSackYardsAsDefense < 0` in `player_play.csv`. Both columns are player-level attributions, meaning they record yardage only for the individual player credited with the sack or half sack. This approach was adopted after finding that the `tackleForALoss` flag, the intuitive alternative, missed a meaningful number of legitimate sacks where the flag was not set despite negative sack yardage being recorded. Using `tackleForALoss` as the sole identifier would have missed 104 half sacks entirely, as `tackleAssist == 1` never co-occurred with `tackleForALoss == 1` in this dataset. Sack totals are computed as:

$$\text{total_sacks} = \text{solo_sacks} + (\text{assisted_sacks} * 0.5)$$

This reflects NFL official scoring convention where half sacks count as 0.5 toward a player's total.

Metrics

The primary metric is Pressure-to-Sack Rate Full (PTSR Full), defined as:

$$\text{PTSR_full} = \text{total_sacks} / \text{total_pressures}$$

Where `total_pressures` is the count of plays where `causedPressure == TRUE` for a given player, and `total_sacks` is the sum of solo sacks plus half sacks weighted at 0.5. `causedPressure` is a PFF-graded binary flag identifying genuine pocket disruption. Players with zero pressures receive `PTSR = 0` to avoid division by zero.

A secondary metric isolating solo finishing rate:

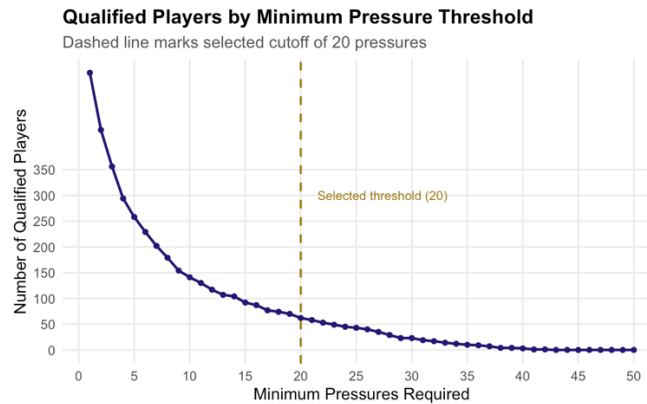
$$\text{PTSR_solo} = \text{solo_sacks} / \text{total_pressures}$$

$$\text{ptsr_gap} = \text{PTSR_full} - \text{PTSR_solo}$$

The independence gap measures the degree to which a player's full PTSR is inflated by half-sacks, where the player was assisted by a teammate in finishing a play. A player at zero finishes entirely independently. A larger gap indicates greater reliance on half sacks.

Pressure rate (`total_pressures / total_pass_rushes`) measures how frequently a player wins his rep. Mean time to pressure (`mean(timeToPressureAsPassRusher)`) measures how quickly he wins when he does. Both are reported alongside PTSR to give evaluators a complete picture of pass rush profile.

A minimum pressure threshold was selected after visual inspection of the qualified player count across all integer thresholds from 1 to 50. The threshold of 20 pressures was selected, yielding 63 qualified players. This cutoff balances sample size reliability against leaderboard depth, at 15 pressures the pool expands to 92 players but includes several with highly unstable rates, while at 25 pressures the pool contracts to 43 players and begins excluding legitimate starters with meaningful production.

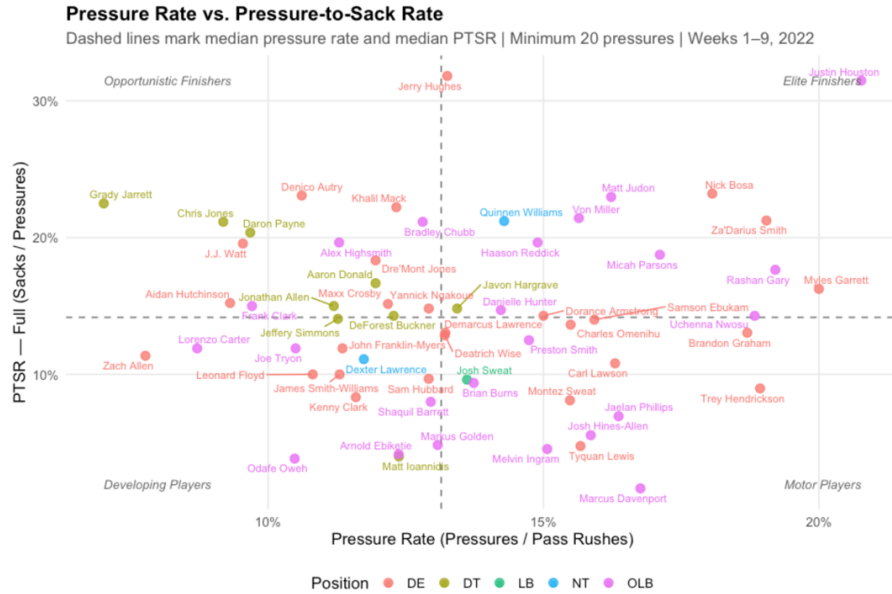


Results

PTSR Full ranges from 1.7% to 31.8% among the 63 qualified pass rushers, with a median of 13.0%. This spread confirms that finishing ability varies substantially across NFL pass rushers, independent of pressure volume. Jerry Hughes (HOU) and Justin Houston (BAL) lead all qualifiers at 31.8% and 31.5% respectively, each converting nearly one in three pressures into sacks. Houston's profile is particularly notable, his 20.8% pressure rate ranks among the highest in the dataset, meaning he wins his rep frequently and finishes at an elite rate when he does.

The leaderboard surfaces two distinct finisher profiles. The first is the high-volume elite finisher, exemplified by Za'Darius Smith (MIN, 19.0% pressure rate, 21.2% PTSR) and Matt Judon (NE, 16.2% pressure rate, 23.0% PTSR), players who generate pressure consistently and convert it efficiently. The second is the opportunistic finisher, exemplified by Jerry Hughes (HOU, 13.3% pressure rate, 31.8% PTSR), players whose pressure volume sits below the median but whose conversion rate is exceptional. PTSR surfaces the second profile in a way that raw sack totals or pressure rate alone cannot.

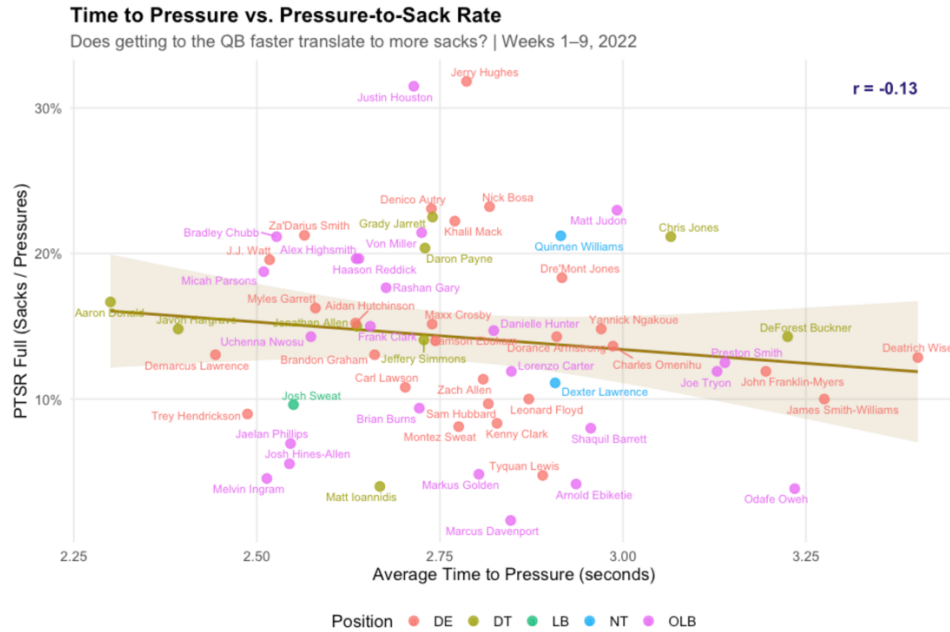
Interior defenders also appear prominently. Grady Jarrett (ATL, DT) and Quinnen Williams (NYJ, NT) rank sixth and tenth respectively, demonstrating that finishing ability is not structurally limited to edge rushers. Jarrett's 7.0% pressure rate reflects the structural disadvantage of his alignment, yet his 22.5% PTSR indicates elite finishing efficiency once he does win his rep.



Time to Pressure Analysis

To evaluate whether finishing ability is related to pass rush speed, a simple linear regression was fitted with mean time to pressure as the explanatory variable and PTSR Full as the response variable. Linear regression was chosen because the relationship between the two variables is hypothesized to be consistent and linear, faster pressures should give the quarterback less time to escape, throw the ball away, or absorb contact and stay upright, producing a straightforward negative relationship between time to pressure and sack conversion rate. The Pearson correlation coefficient was computed alongside the regression to quantify the strength and direction of that relationship independently of the regression slope, providing a single interpretable summary statistic for the chart.

The Pearson correlation between mean time to pressure and PTSR Full is $r = -0.13$ across the 63 qualified players. The negative direction confirms the intuitive hypothesis, players who reach the quarterback more quickly tend to convert at modestly higher rates. However, the magnitude indicates that time to pressure explains very little of the variation in finishing ability. Two players illustrate the limits of this relationship in opposite directions. Matt Judon (NE) generates pressure in nearly 3 seconds on average, among the slowest in the dataset, yet converts at 23%. Trey Hendrickson (CIN) reaches the quarterback in 2.48 seconds and generates pressure on 18.9% of his snaps, both above-average marks, yet converts at only 9%. The data suggests that what happens after a defender reaches the quarterback, leverage, redirection, and finishing technique, is a more significant driver of sack conversion than speed of arrival.



Limitations

causedPressure is a PFF-graded tag applied by human analysts. This introduces inter-rater variability into the denominator of every PTSR calculation. Two players with identical on-field performance could receive different pressure counts depending on grading decisions, particularly on plays involving stunts, twists, or shared credit situations. Any comparison of PTSR values across players should account for this source of noise.

PTSR does not control for the quality of offensive linemen faced. A pass rusher who generates pressure and sacks primarily against below-average tackles will post higher PTSR values than an equally skilled rusher facing elite protection. Without opponent adjustment, PTSR reflects a combination of finishing skill and schedule difficulty that cannot be fully disentangled in this dataset.

Nine weeks of data produces stable estimates for high-volume pass rushers but introduces meaningful variance for players near the 20-pressure threshold. A player with 20 pressures and 4 sacks has a PTSR of 20.0%, but a single sack in either direction shifts that figure by 5 percentage points. Results for players near the threshold should be interpreted with appropriate caution.

The solo vs. assisted split partially addresses this, but PTSR does not distinguish between one-on-one pass rush situations and stunt or twist designs. A player who primarily finishes as the cleanup rusher on a well-designed twist is a different evaluative profile than one who wins individual battles. Play-level blocking assignment data would be required to make this distinction cleanly.

Coverage type influences how quickly a quarterback releases the ball, which in turn affects whether a pressure becomes a sack opportunity. A pass rusher operating behind a man coverage shell faces a quarterback who must hold the ball longer than one operating behind a zone shell with quick outlet routes. `pff_manZone` is available in `plays.csv` and could be used to stratify PTSR by coverage type in a future iteration.

The BDB dataset does not capture the full 2022 season, and individual play-level gaps exist within the nine-week window. Jerry Hughes provides a documented example. Seven sacks are recorded for him in the dataset against eight confirmed by official NFL statistics. Comparisons to publicly reported sack totals should account for this coverage limitation.

Future Extensions

The most valuable extension would be controlling for offensive line quality faced. Pass rushers could be adjusted using a blocker quality metric derived from the same dataset, for example the pressure rate allowed by each individual offensive lineman, to produce an opponent-adjusted PTSR that isolates finishing skill more cleanly from schedule effects.

Splitting PTSR by down and distance would add meaningful context. A pass rusher who converts at a high rate on third and long, when the quarterback has maximum time to wait for a receiver, is demonstrating a different and arguably more valuable skill than one whose sacks come primarily on early downs with quick release concepts.

The logical extension of PTSR is a play-level model that estimates the probability of a sack given a pressure event, incorporating time to pressure, coverage type, down and distance, and blocker identity as features. This would move from a descriptive rate to a predictive framework, enabling evaluation of individual plays rather than season-long aggregates.

A single nine-week sample is sufficient to identify finishing ability but insufficient to assess how stable PTSR is year over year. Running the same analysis across multiple seasons and computing year-to-year correlations would establish whether PTSR is a repeatable skill or a noisy short-sample artifact. High year-to-year correlation would significantly strengthen the case for using PTSR as an evaluative tool in player personnel decisions.

Incorporating the `blockedPlayerNFLId` columns available in `player_play.csv` would allow PTSR to be stratified by whether a pressure was generated in a one-on-one situation versus a stunt or twist design. This would produce a cleaner measure of individual pass rush finishing skill by isolating plays where the rusher was solely responsible for both winning the rep and finishing the play.